Year 7 Astronomy – Solar System Name: _____

1. Which planet is very similar in size to our Earth? What is the difference in their diameter?

2. Which planet has the biggest mass?

3. Which planet (Venus or Mars) is closer to our Earth?

4. Why does it take much longer time for Neptune to go around (revolve/ orbit) the Sun than for Mercury or our Earth? (2 REASONS …. Find the answers from the data)

5. Which planet has almost the same hours in a day compared to the Earth?

6. Calculate how many hours are there in a day in Neptune.
   Given an example – The rotation period is the hours of a day
   In Jupiter – the rotation period is only 0.41 of the Earth = 0.41 x 24 = 9.8 hours only.

7. Why do the outer planets such as Saturn and Neptune are so cold?

8. Mercury is the closest to the Sun, yet it has a very low temperature at night? What is the lowest temperature of Mercury? Give a reason for having such a low temperature.
   (Hint: data from atmospheric composition)

9. Which is the hottest planet? Find out why?

10. On which planet do you weigh the most? Why?

11. Which planet has the largest number of moons? Give a reason why it can have such a large number.

12. Give some differences between the Inner Planets and Outer Planets.
    (NB: the first 4 are the inner, and the last 4 are the outer planets)

	Inner Planets	Outer Planets
They are:		
Surface:		
Distance:		
Rings:		
Moons:		

13. Which planet will float on water?

Planetary Fact Sheet - Metric

	MERCURY	VENUS	EARTH	MOON	MARS	JUPITER	SATURN	URANUS	NEPTUNE	PLUTO
Mass (10^{24}kg)	0.330	4.87	5.97	0.073	0.642	1898	568	86.8	102	0.0131
Diameter (km)	4879	12,104	12,756	3475	6792	142,984	120,536	51,118	49,528	2390
Density (kg/m^3)	5427	5243	5514	3340	3933	1326	687	1271	1638	1830
Gravity (m/s^2)	3.7	8.9	9.8	1.6	3.7	23.1	9.0	8.7	11.0	0.6
Escape Velocity (km/s)	4.3	10.4	11.2	2.4	5.0	59.5	35.5	21.3	23.5	1.1
Rotation Period (hours)	1407.6	-5832.5	23.9	655.7	24.6	9.9	10.7	-17.2	16.1	-153.3
Length of Day (hours)	4222.6	2802.0	24.0	708.7	24.7	9.9	10.7	17.2	16.1	153.3
Distance from Sun (10^6 km)	57.9	108.2	149.6	0.384*	227.9	778.6	1433.5	2872.5	4495.1	5870.0
Perihelion (10^6 km)	46.0	107.5	147.1	0.363*	206.6	740.5	1352.6	2741.3	4444.5	4435.0
Aphelion (10^6 km)	69.8	108.9	152.1	0.406*	249.2	816.6	1514.5	3003.6	4545.7	7304.3
Orbital Period (days)	88.0	224.7	365.2	27.3	687.0	4331	10,747	30,589	59,800	90,588
Orbital Velocity (km/s)	47.9	35.0	29.8	1.0	24.1	13.1	9.7	6.8	5.4	4.7
Orbital Inclination (degrees)	7.0	3.4	0.0	5.1	1.9	1.3	2.5	0.8	1.8	17.2
Orbital Eccentricity	0.205	0.007	0.017	0.055	0.094	0.049	0.057	0.046	0.011	0.244
Axial Tilt (degrees)	0.01	177.4	23.4	6.7	25.2	3.1	26.7	97.8	28.3	122.5
Mean Temperature (C)	167	464	15	-20	-65	-110	-140	-195	-200	-225
Surface Pressure (bars)	0	92	1	0	0.01	Unknown*	Unknown*	Unknown*	Unknown*	0
Number of Moons	0	0	1	0	2	67	62	27	14	5
Ring System?	No	No	No	No	No	Yes	Yes	Yes	Yes	No
Global Magnetic Field?	Yes	No	Yes	No	No	Yes	Yes	Yes	Yes	Unknown
	MERCURY	VENUS	EARTH	MOON	MARS	JUPITER	SATURN	URANUS	NEPTUNE	PLUTO